

OCTriage: which OCT segmentations should graders read first?

Two Flywheel gears that tell graders what to read first: an independent case study for the Wisconsin Reading Center by Kaan Mutlu

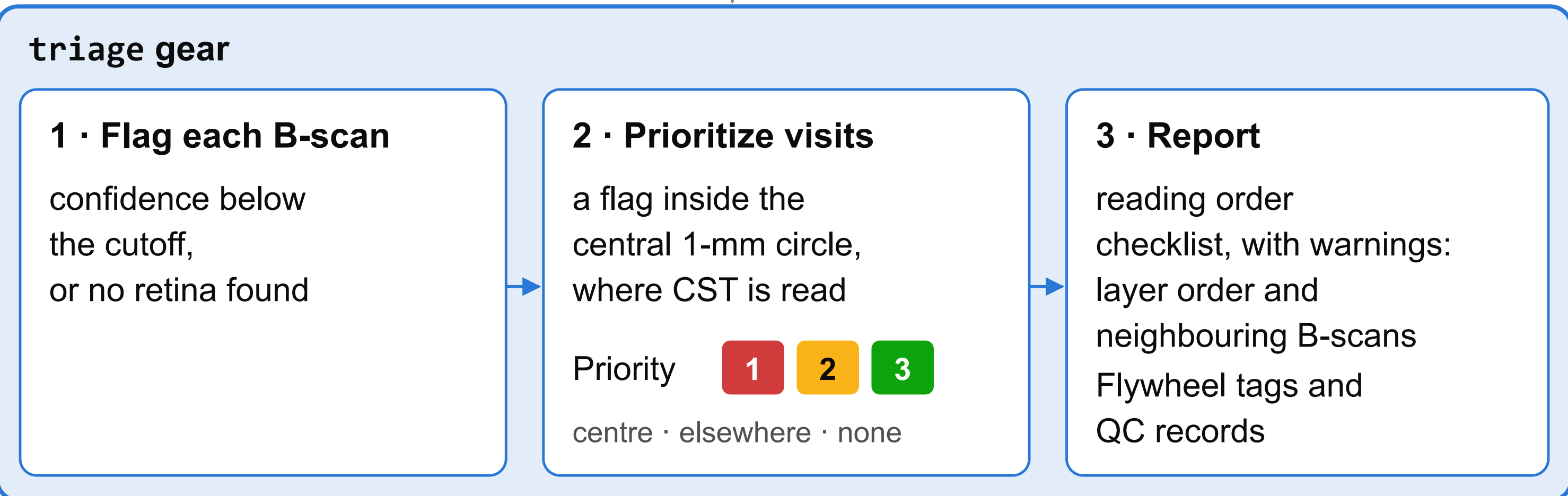
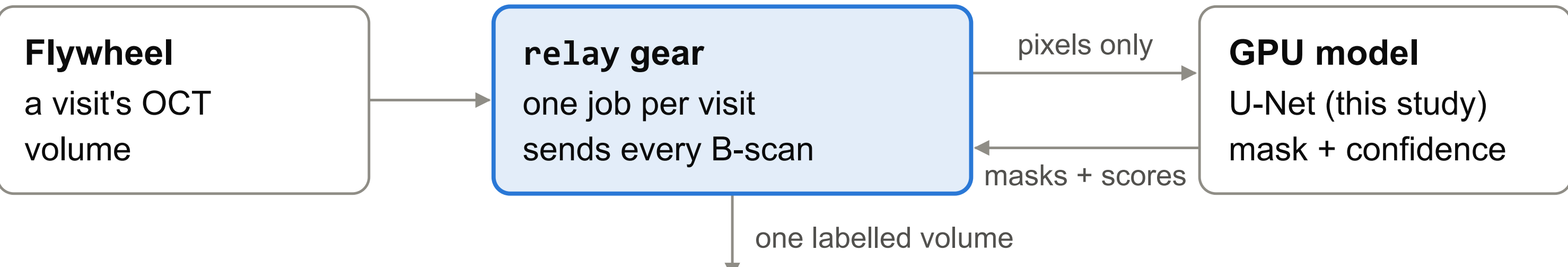
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Why it matters

At WRC, graders review what AI does,¹ although most masks need little or no editing. **OCTriage tells them which to read first, steering their effort toward likely errors.**

- In WRC's GA study, graders reviewed every AI mask and found 84.8% needed no or minor edits.²
- Readers tire: after a day of reading, radiologists found fractures less accurately, their eyes focused less well, and they felt more tired.³
- In 2022 WRC used an AI score and a cutoff to send only likely label errors to graders, and estimated grader time at 25% of a full pass.⁴ OCTriage uses the same idea, but every scan is still read; only the order changes.
- Review boards limit where patient data may go, and clinicians won't use a tool that starts at a command prompt.⁵ A Flywheel gear can call a GPU server, and WRC's GPUs sit inside UW;⁶ OCTriage's relay gear is built to carry images along that route.

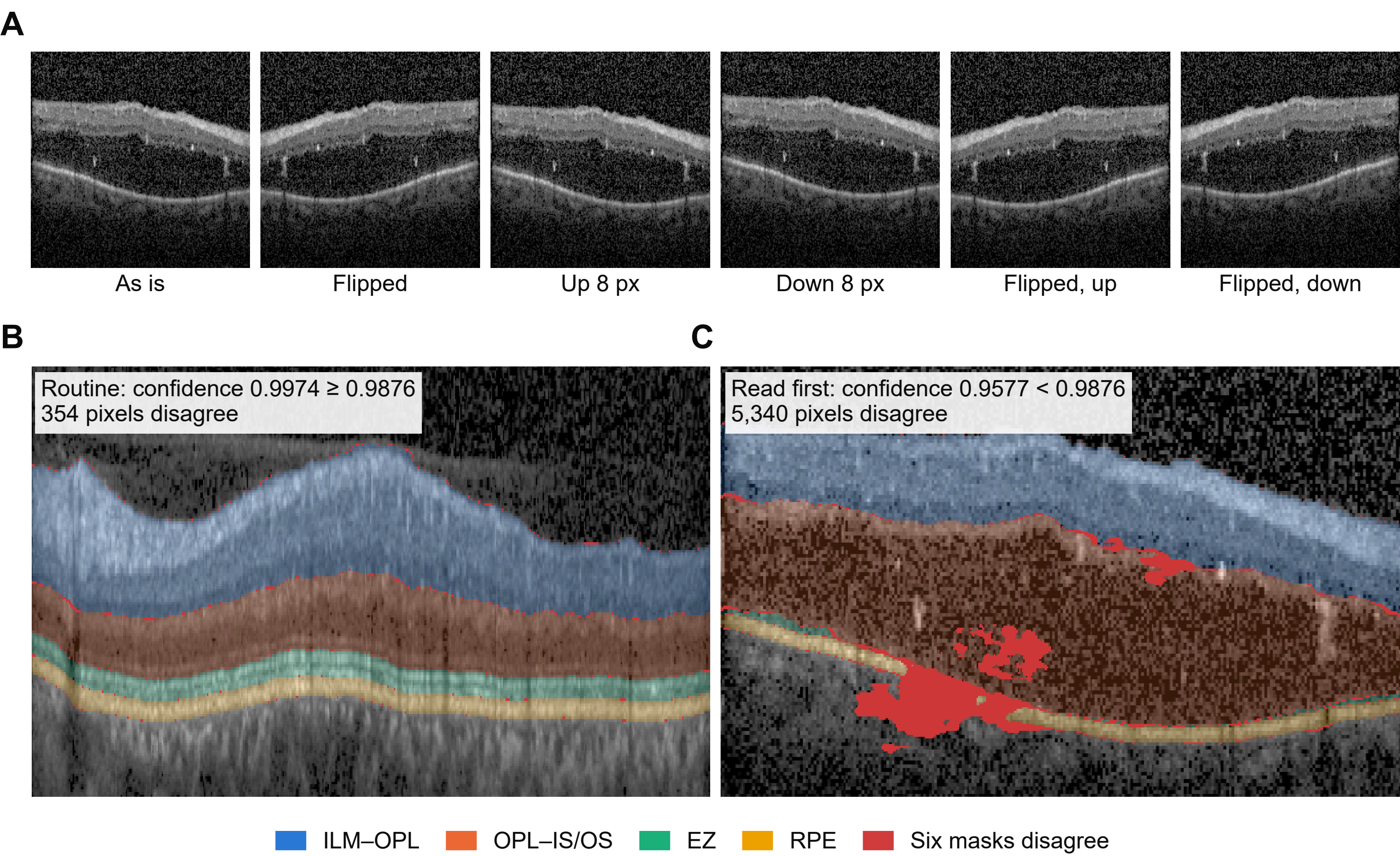
What I built



- Two gears with no model inside: reLay sends each visit's B-scans to a GPU model and stores the masks and confidence; triage turns them into a reading order. Any model that returns a mask and a confidence fits, WRC's nnSAM⁷ included.
- I trained a 2D U-Net with 5-fold cross-validation by patient for this study.

How confidence is measured

Six altered copies of each B-scan are segmented (test-time augmentation⁸); confidence is how well their masks agree. The cutoff, 0.9876, is Youden's index:⁹ the best balance between poor masks caught and acceptable masks flagged. In testing it was chosen on four fifths of the patients and applied to the other fifth.



A The six copies. **B** Routine: the masks agree. **C** Read first: they disagree (red); confidence is below the cutoff.

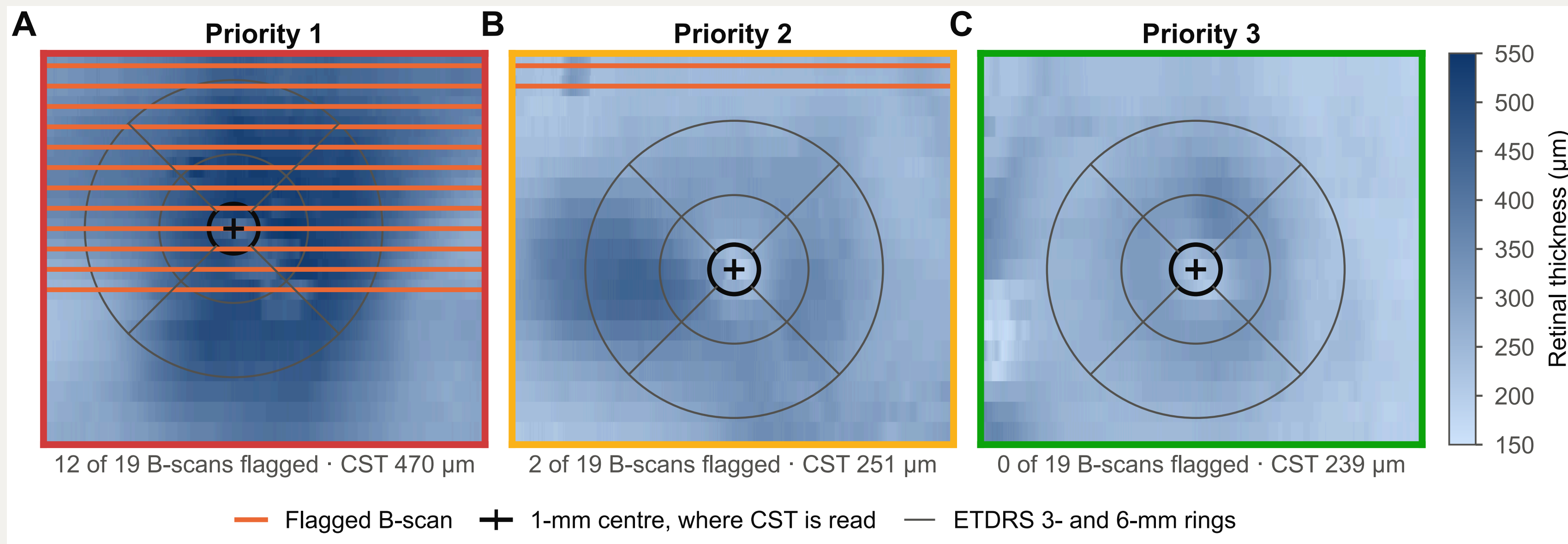
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The reading order

- Priority 1:** a flagged B-scan crosses the 1-mm centre where central subfield thickness (CST) is read, or the scan is off-centre. CST steers trials: in a major DME drug trial (DRCR Protocol T), it helped decide who could join and when an eye got another injection.¹⁰
- Priority 2:** flagged B-scans elsewhere. A B-scan is flagged when its confidence is below the cutoff or no retina is found.
- Priority 3:** nothing flagged.

Layer-order and neighbouring-B-scan checks appear on the checklist as warnings.



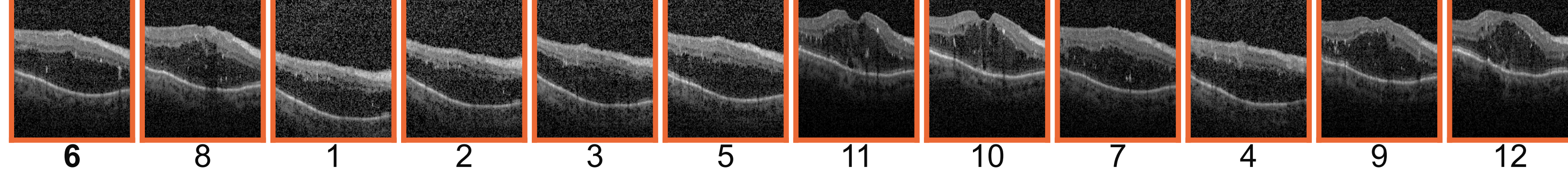
Thickness maps of three DME visits, one per priority, with the ETDRS grid centred on the fovea, the centre of vision (+).¹¹ Orange lines: flagged B-scans.

What a grader would see in Flywheel

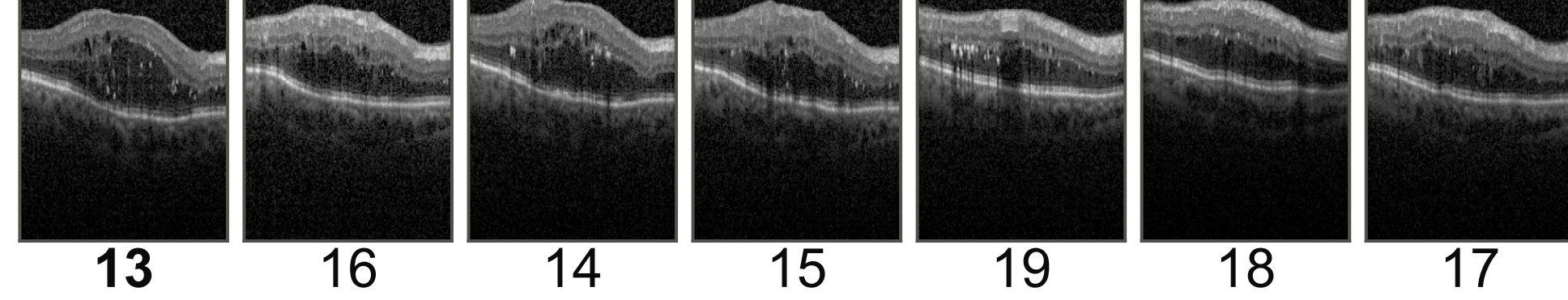
A Priority	Patient	Flagged B-scans	CST (μm)
Priority 1 — endpoint at risk	DME-07	19 of 19	440
Priority 1 — endpoint at risk	DME-06	18 of 19	230
Priority 1 — endpoint at risk	AMD-26	14 of 61	258
Priority 1 — endpoint at risk	DME-28	13 of 19	238
Priority 1 — endpoint at risk	DME-11	12 of 19	470
	:		
Priority 2 — flags outside the centre	AMD-21	4 of 61	213
Priority 2 — flags outside the centre	DME-14	2 of 19	251
	:		
Priority 3 — no flags	AMD-08	0 of 19	219

B Patient DME-11 · Priority 1: flagged B-scans 8-10 cross the central 1-mm circle, where CST is measured

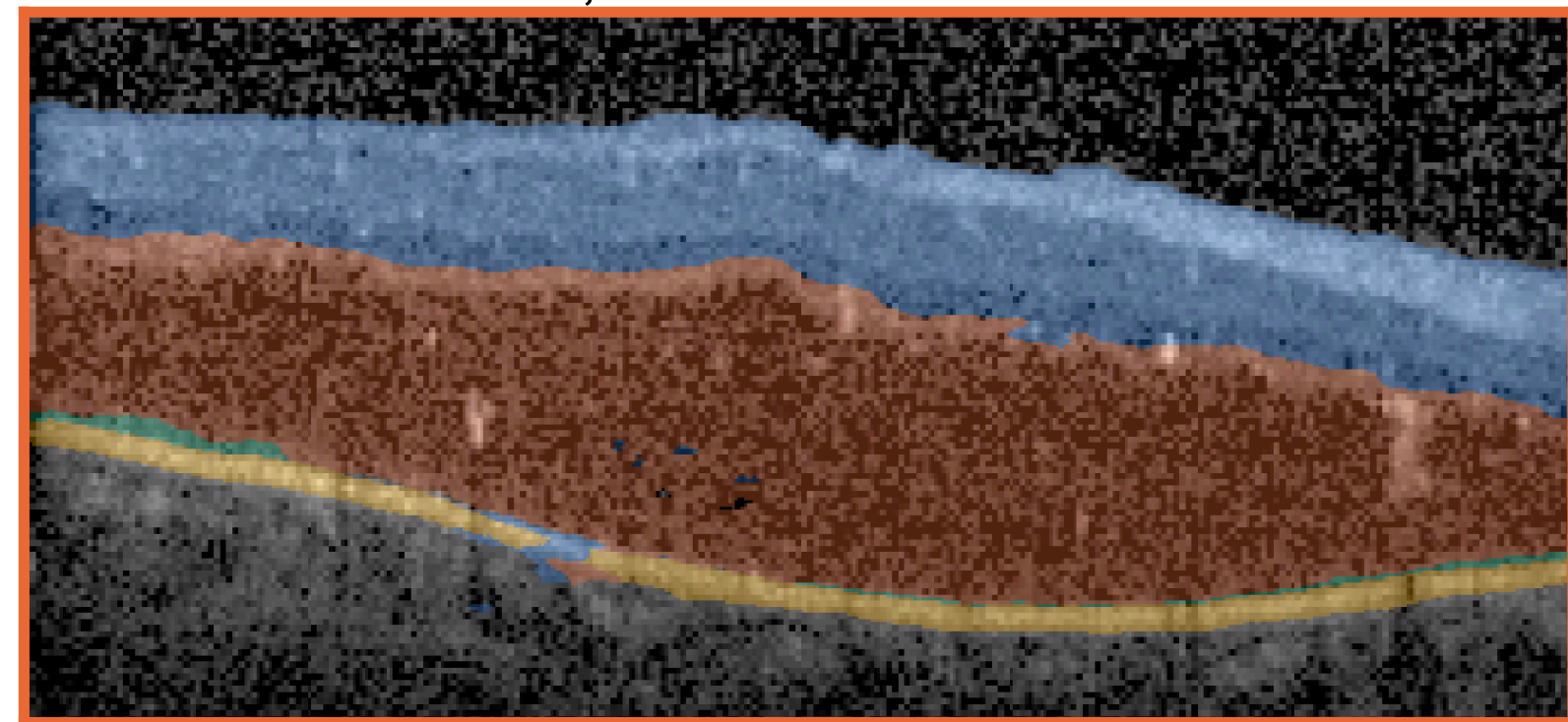
Read first: the 12 flagged, least confident first



Then the other 7

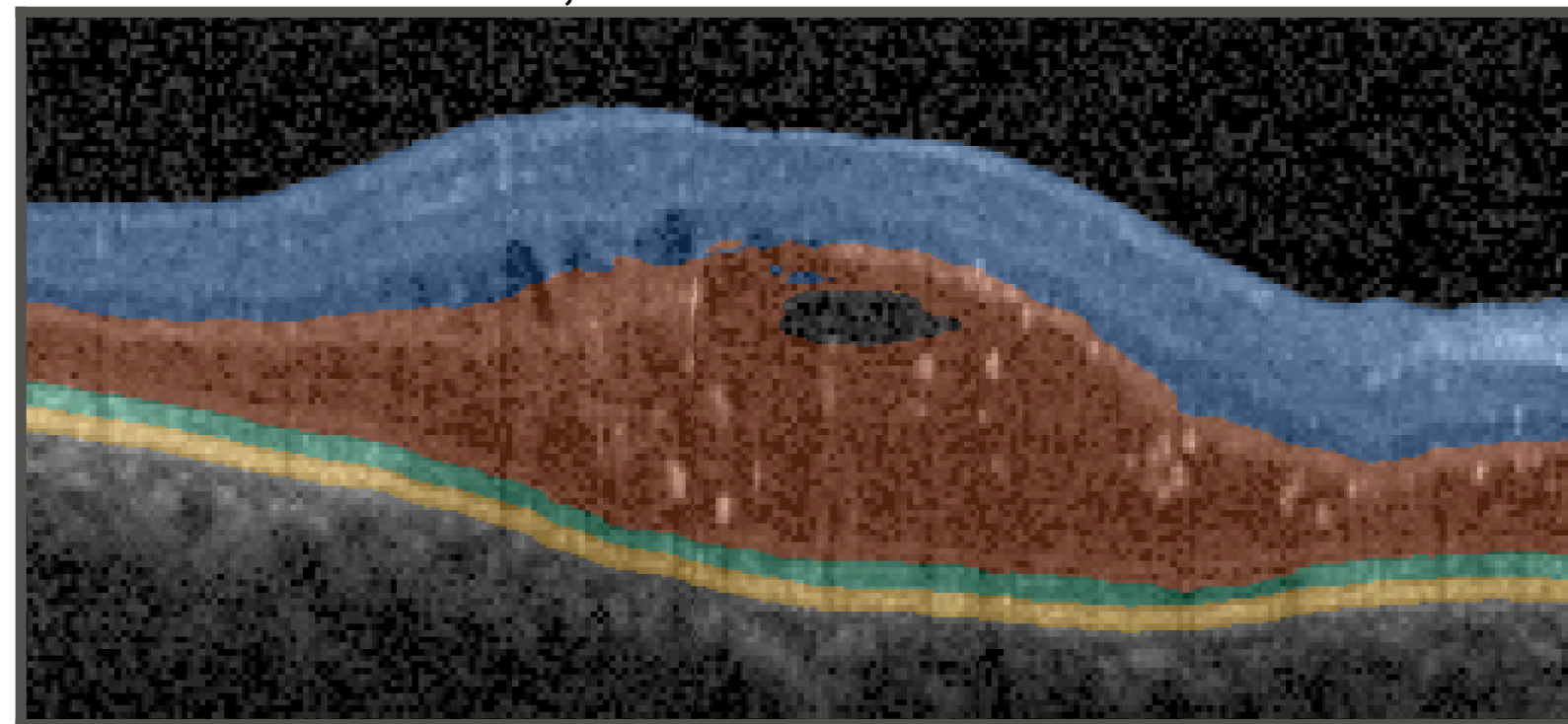


C B-scan 6: read first, 1st in order



- ✗ confidence 0.958 < cutoff 0.988
- ✓ retina found
- ✗ layer order broken (warning), -1.3 to +1.9 mm from the fovea
- ✓ consistent with its neighbouring B-scans
- EZ not seen in 41%: loss or error?

B-scan 13: routine, 13th in order



- ✓ confidence 0.992 >= cutoff 0.988
- ✓ retina found
- ✗ layer order broken (warning), +0.5 to +1.6 mm from the fovea
- ✓ consistent with its neighbouring B-scans

A The workflow. **B** Patient DME-11's B-scans in reading order. **C** A read-first B-scan beside a routine one, each with its checklist. On trial data the gear would also check image quality and change since the last visit (longitudinal consistency);¹² OCT5k has no quality grades and one visit per patient, so neither check could be tested here.

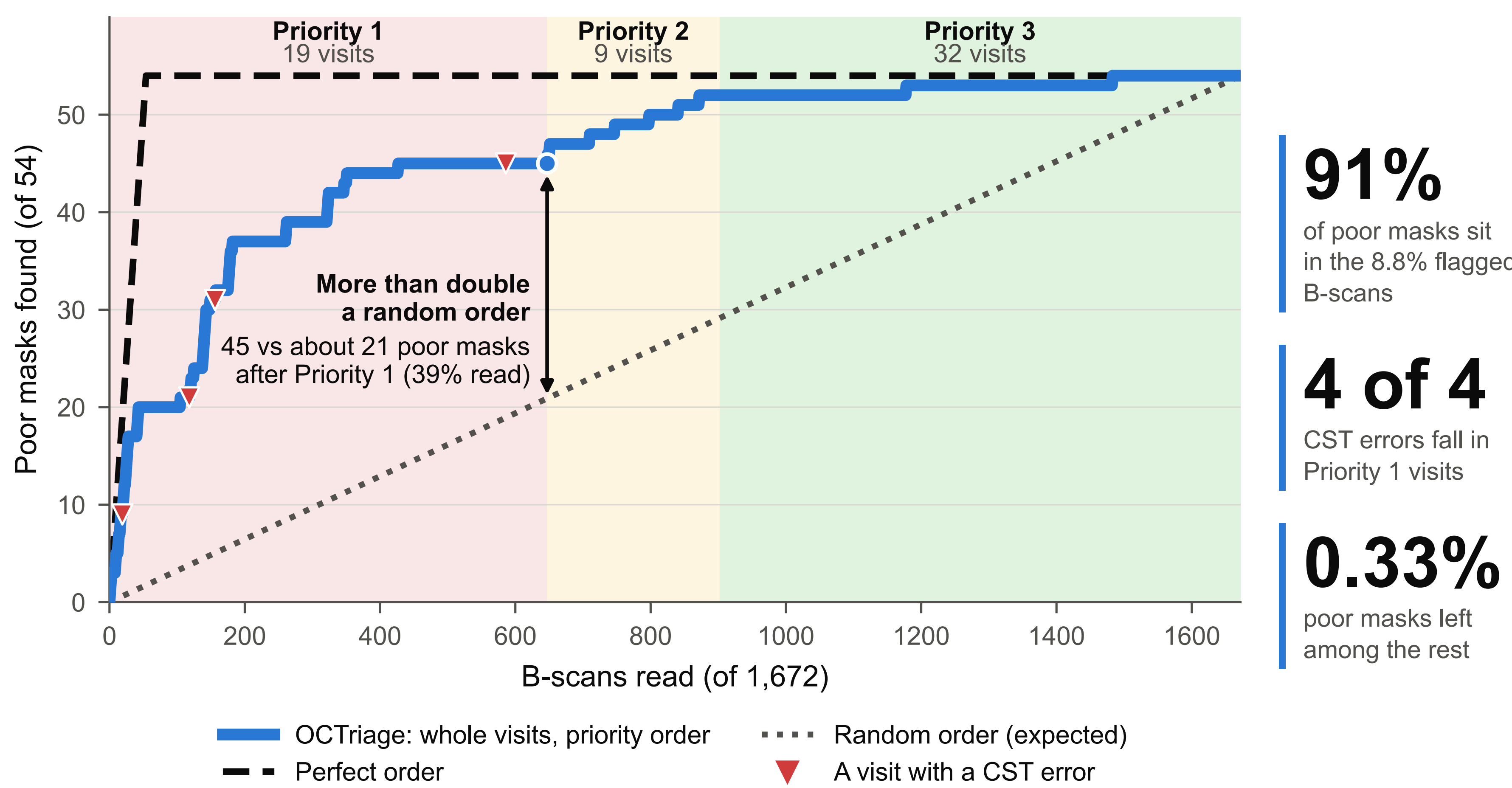
Data: OCT5k, three graders per B-scan

1,672 B-scans from 60 patients (AMD, DME, healthy), public (CC0).¹² OCT5k stores each B-scan as its own image; I stacked each patient's B-scans in order into one volume, spaced by the paper's scan size (8.9 × 7.4 mm). Answer key: a mask is poor if it agrees with the graders at least 0.05 Dice less than they agree with each other.^{12,13}

Cohort	Patients	B-scans	Grader vs grader Dice	Model vs graders Dice	Poor masks
AMD	21	739	0.916	0.919	21
DME	19	403	0.892	0.892	24
Normal	20	530	0.926	0.932	9
All	60	1,672	0.914	0.917	54

What it shows

Read visit by visit in priority order, the Priority 1 visits (39% of B-scans) hold 45 of the 54 poor masks and all 4 CST errors; a random order finds about 21.

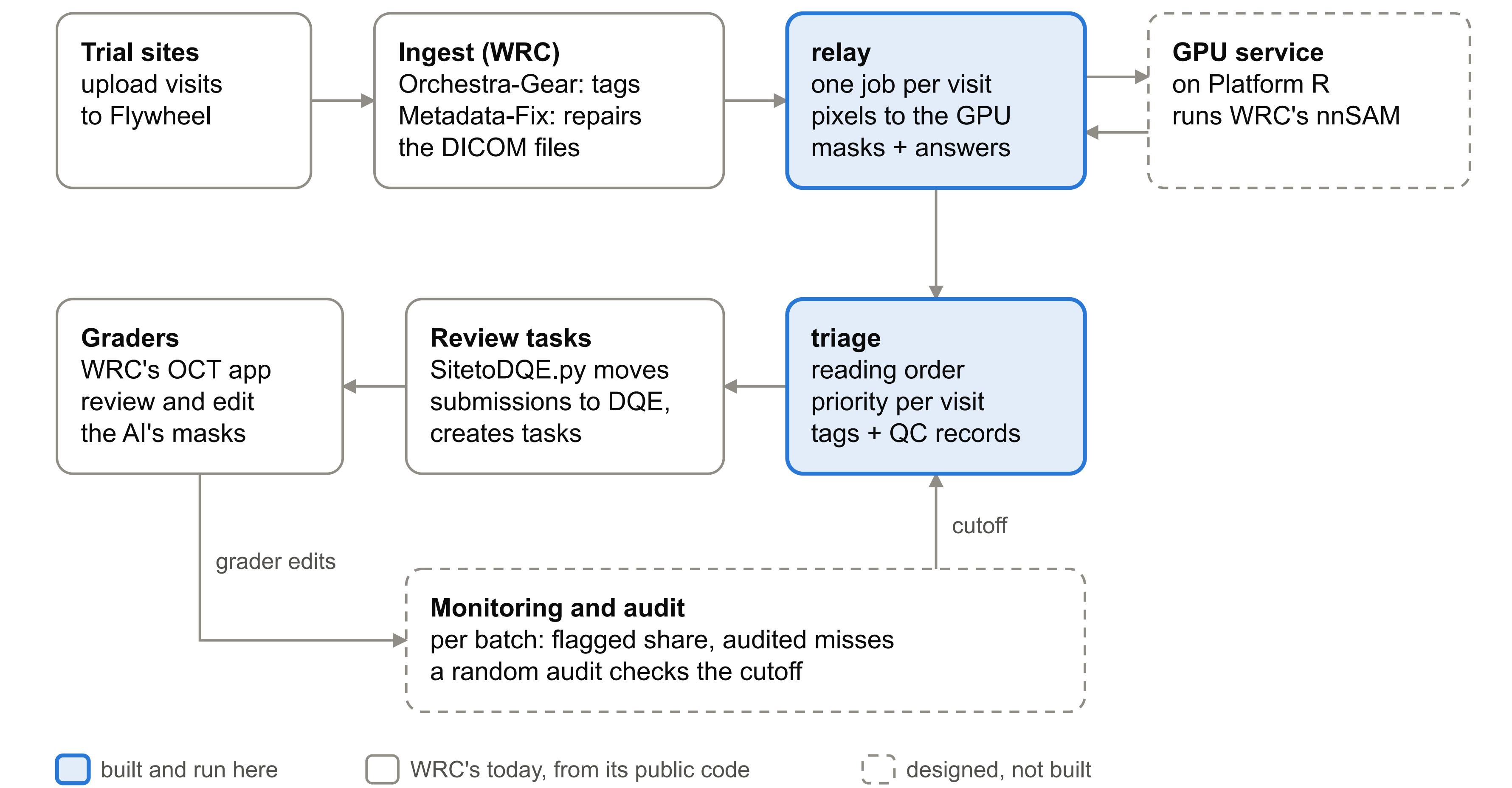


91%
of poor masks sit in the 8.8% flagged B-scans

4 of 4
CST errors fall in Priority 1 visits

0.33%
poor masks left among the rest

How it would fit at WRC



built and run here WRC's today, from its public code designed, not built

Both gears run at ingest, after WRC's Orchestra-Gear and Flywheel-Metadata-Fix and before SitetoDQE.py moves site submissions into a review (DQE) project and creates tasks (M. Younis¹⁴). The placement is my interpretation of WRC's public code, not confirmed with WRC.

Conclusion

WRC's graders review what AI does,¹ and for trial endpoints that is likely to stay: an OCT measurement was the primary endpoint behind a 2025 drug approval,¹⁵ and FDA guidance on AI in drug development is still a draft.¹⁶ OCTriage keeps full reading and changes only the order: the first 39% of scans held 83% of the poor masks, more than double a random order, so graders meet likely errors first. reLay works with any model behind the same interface; triage's OCT rules sit in one module, so other imaging would add its own.

Limitations: one public dataset; 4 endpoint errors; one model and one device; no live Flywheel site; sparse volumes.